

Target Vocabulary – Week 3

New senses for humans

Perception n. /pə'sep.ʃən/	a belief or opinion, often held by many people and based on how things seem:	We have to change the public's perception that money is being wasted. These photographs will affect people's perceptions of war.
Electromagnetic adj. /i,lek.trəʊ.mæg 'net.ɪk/	relating to the electrical and magnetic forces produced by an electric current	An electromagnetic wave is created as a result of the interaction between an electric field and a magnetic field.
Receptor n. /rɪ'sep.tər/	a nerve ending that reacts to a change, such as heat or cold, in the body by sending a message to the central nervous system	Serotonin receptors are classified into seven different families depending on their signalling pathways. Further, one would also predict that these regions ultimately receive input from deep receptors in the muscles and joints.
Infrared adj. /,ɪn.frə'red/	Infrared light is a type of light that feels warm but cannot be seen	<i>Their pilots are guided by an infrared optical system that shows images clearly even at night.</i>
Ultraviolet adj. /,ʌl·trə'veɪ·ə·lɪt	Ultraviolet light has a wavelength that is after the violet (= light purple) end of the range of colours that can be seen by humans. Light of this type causes the skin to become darker in the sun.	Fragments were separated through electrophoresis on 2 % agarose gels and visualized using ethidium bromide under ultraviolet light.
Electrochemical adj. /i,lek.trəʊ 'kem.ɪ.kəl/	relating to the production of chemical changes using electricity, or the electricity produced by chemical c	All samples were analysed with a pulsed electrochemical detector and peak heights measured with internal standardization.

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Lavishly adj. /'læv.ɪʃ/	large in quantity and expensive or impressive: lavish gifts/promises/praise	The evening was a lavish affair with glorious food and an endless supply of champagne.
Echolocation n. /ˌek.əʊ.ləʃən/ 'keɪ.ʃən/	a process in which animals, for example bats (= small animals with wings that fly at night), find their way in the dark by producing sound waves that echo (= are heard again) when they are reflected off an object:	Dolphins locate underwater creatures using echolocation . In the domain of vision, two main classes of such sensory substitution devices have been constructed: echolocation devices and tactile visual substitution devices.
Appendage n. /ə'pen.dɪdʒ/	something that exists as a smaller and less important part of something larger:	The committee is a mere appendage of the council and has no power of its own.
Gauge v. /geɪdʒ/	to calculate an amount, especially by using a measuring device	Use thermometer to gauge the temperature.
Multidimensional adj. /ˌmʌl.tɪ.daɪ'men.ʃən.əl/	having many different <u>features</u>	Crime is a multidimensional problem.
Machinery n. /mə'ʃiː.nər.i/	a group of large machines or the parts of a machine that make it work	The noise of the machinery made her voice inaudible.
Game changer n. /'geɪmˌtʃeɪn.dʒər/	something or someone that affects the result of a game very much something such as a product or event that affects a situation or area of business very much	As a player he can be a game changer .
Inherently adv. /ɪn'her.ənt.li/ /ɪn'hɪə.rənt.li/	in a way that exists as a natural or basic part of something	Thanks to their inherently larger bodies, men typically possess larger brains than women do.
Impoverished	The word impoverished can be used	Benefits should be targeted at the

adj. <i>/ɪmˈpɒv.ər.ɪft/</i>	to describe a person or area that is extremely poor.	most impoverished families.
Umwelt n. <i>/ˈʊmvɛlt/</i>	the environmental factors, collectively, that are capable of affecting the behaviour of an animal or individual	its best it creates its own umwelt , a parallel world recognizable but distinct.
Magnetite n. <i>/ˈmæɡniˌtaɪt/</i>	A black magnetic mineral, found in igneous and metamorphic rocks and as a separate deposit. It is a source of iron. Composition: iron oxide. Formula: Fe ₃ O ₄ . Crystal structure: cubic	This works well, although is a little unclear when a 'black' mineral like magnetite is viewed against the black background.
Sensory adj. <i>/ˈsen.sər.i/</i>	connected with the physical senses of touch, smell, taste, hearing, and sight	First, the representations we create are limited by sensory constraints.
Substitution n. <i>/ˌsʌb.stɪˈtjuːʃən/</i>	the use of one person or thing instead of another	The only substitution I made was to use oat milk instead of almond milk.
X-ray n. <i>/ˈeks.reɪ/</i>	a type of radiation that can go through many solid substances, allowing hidden objects such as bones and organs in the body to be photographed:	A CT scan uses X-rays to take a picture of everything under the skin.
Consciousness-raising n. <i>/ˈkɒn.ʃəs.nəsˌreɪ.zɪŋ/</i>	the attempt to increase people's knowledge of and interest in social and political matters	The strategy instruction was generally explicit and students' metacognition was developed through a variety of consciousness-raising activities.
Real-time adj. <i>/ˈriəl.taɪm/</i>	communicated, shown, presented, etc. at the same time as events actually happen	real-time computer systems (= processing data immediately as it comes in)

Biomechatronics

Haptic adj. <i>/ˈhæptɪk/</i>	relating to or involving the sense of touch	Players use a haptic device such as a joystick to control the game.
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Haptics n.	the study or use of tactile sensations and the sense of touch as a method of interacting with computers and electronic devices	Haptics is defined as a technology that transmits tactile information using sensations such as vibration, touch, and force feedback.
Sensation n. <i>/sen'sei.ʃən/</i>	1. a feeling that you get when something affects your body 2. the ability to feel through your sense of touch	A sensation of burning or tingling may be experienced in the hands. She seemed to have lost all sensation in her arms.
Tactile adj <i>/'tæktail/</i>	connected with the sense of touch; using your sense of touch	visual and tactile communication
Exoskeleton n. <i>/'eksəʊskelɪtn/</i>	a machine attached to somebody's body to enable them to perform movements and actions more easily	Robotic exoskeletons add strength to healthy limbs and restore movement to weak ones.
Demarcate v. <i>/'di:ma:kert/</i>	to mark or establish the limits of something	The police demarcated the city into eighteen geographical divisions.
Mimetic adj <i>/mi'metɪk/</i>	copying the behaviour or appearance of somebody/something else	Mimetic art/literature (= that copies real life)
Biomimetics	the study and development of synthetic systems that mimic the formation, function, or structure of biologically produced substances and materials and biological mechanisms and processes.	Humans have been trying to learn biomimetics for centuries by mimicking nature and its behaviors and processes in order to develop novel materials, structures, devices, and technologies.
Biomechatronics	an applied interdisciplinary science that aims to integrate biology and mechatronics	Biomechatronics is the development and optimization of mechatronic systems using biological and medical knowledge.
Mechatronics n. <i>/,mekə'trɒnɪks/</i>	technology that combines electronics and mechanical engineering	Mechatronics is a multi-faceted field that combines several types of engineering (mechanical, computer, and electrical) with machining, robotics, and information technology.
Retina n. <i>/'retɪnə/</i>	a layer of tissue at the back of the eye that is sensitive to light and sends signals to the brain about what is seen	The treatment comes in the form of an injection into the space behind the retina .

Relay v. <i>/ˈriːleɪ/</i>	to receive and send on information, news, etc. to somebody	Instructions were relayed to him by phone.
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